COSC 4337 Group 11 Project Proposal:
Using ML to Classify Animal Species by Their Sounds

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Dataset:

- BirdCLEF+ 2025 (Audio Bird/Animal Classification) - [BirdCLEF+ 2025 | Kaggle]
  - Dataset: The training dataset includes

    primary_label: A code for the species (eBird code for birds, iNaturalist taxon ID for non-birds). You can review detailed information about the species by appending codes to eBird and iNaturalis taxon URL, such as https://ebird.org/species/gretin1 for the Great Tinamou or https://www.inaturalist.org/taxa/24322 for the Red Snouted Tree Frog. Not all species have their own pages; some links might fail.

    secondary_labels: List of species labels that have been marked by recordists to also occur in the recording. Can be incomplete.

    latitude & longitude: Coordinates for where the recording was taken. Some bird species may have local call 'dialects,' so you may want to seek geographic diversity in your training data.

    author: The user who provided the recording. Unknown if no name was provided.

    filename: The name of the associated audio file.

    rating: Values in 1..5 (1 - low quality, 5 - high quality) provided by users of Xeno-canto; 0 implies no rating is available; iNaturalist and the CSA do not provide quality ratings.

    collection: Either XC, iNat or CSA, indicating which collection the recording was taken from. Filenames also reference the collection and the ID within that collection.

Introduction:

- We plan to build a deep learning system that identifies multiple animal species (birds, mammals, amphibians, and insects) from audio recordings collected in the Middle Magdalena Valley of Colombia. Our approaches will include:
  - **Data Preprocessing**: Prepare the audio data by converting the recordings into spectrogram representations. Split long recordings into smaller time segments for model input. Normalize audio signals across recordings and apply noise reduction techniques to reduce background noise. Examine which species have abundant vs scarce training examples to plan strategies for handling imbalance data. We may use preprocessing methods, such as data augmentation, to improve performance.
  - **Training model**: Build and train CNNs on spectrogram images. We may test different model variations (such as pretrained models) and compare their performance. Since some species may have few recordings, we will experiment with helping the model learn rare classes better. To address class imbalance, we will experiment with techniques such as class-weighted loss functions or oversampling of rare species during training to increase the diversity of rare species examples. Tune hyperparameters for better performance.
  - **Evaluation**: Measure performance using metrics such as macro F1-score, precision, and recall. Analyze per-class performance and rare species performance. Generate confusion matrix to identify common misclassifications. Compare the different approaches. Perform error analysis to understand model limitations.

- **Briefly describe related work (if applicable).**

There has been extensive research in bioacoustics, especially in species identification using audio recordings. Some research includes the "Animal Species Classifications from vocalizations using cochlear inspired features and Machine Learning" by Karim Youseff, this research explores using cochlear implant voice filtering technology to reduce noise, and to distinguish sound frequencies by mimicking the natural process of the human auditory system.

The cochlea is a coiled tube in the human ear that separates incoming sound into frequencies using sensitivity to the frequencies along the tube. The researchers replicated this idea of a filtering process to extract acoustic features for classifying animal species.

Our approach will compare classification accuracy using the spectrogram features and compare it to the cochlear inspired features to find which best enhances frequency discrimination in animal sounds.

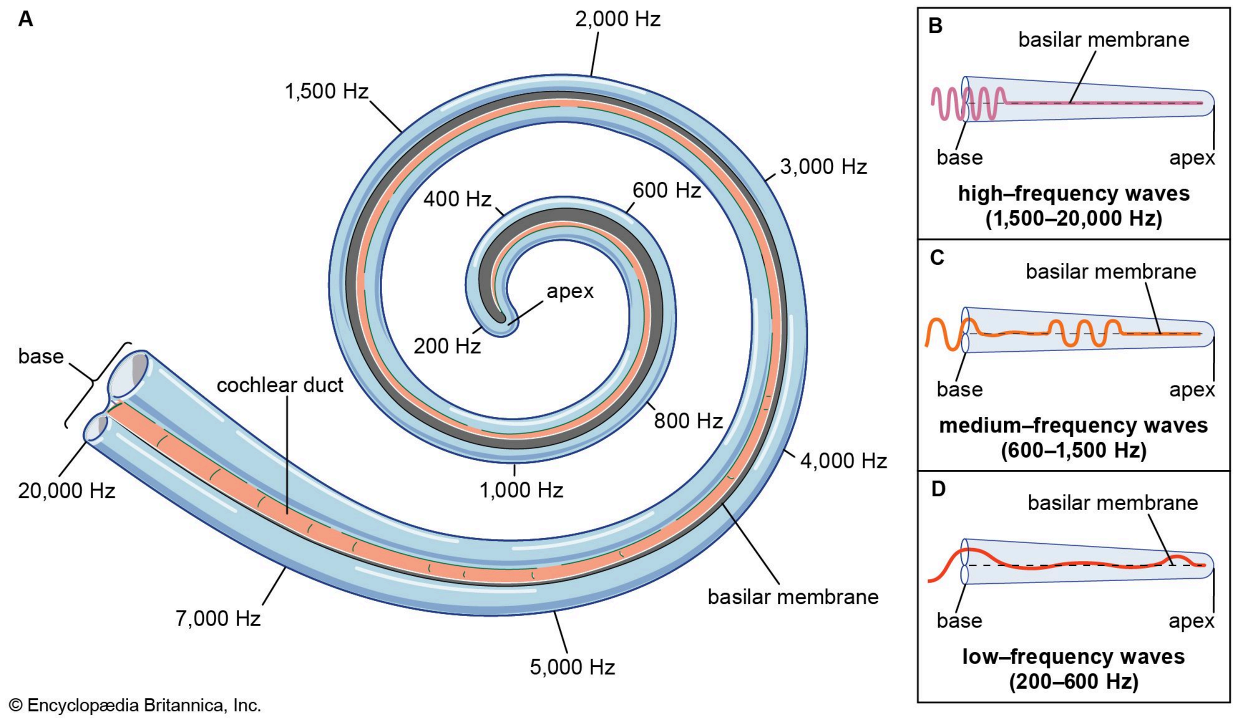

Fig. 1 Human ear cochlea representation. As the sound travels through the basilar membrane, the cochlea picks up on the frequency, with the base detecting decreasing frequencies as the sound travels towards the apex.

Motivation:

- **Describe why your project is E.g., you can describe why your project could have a broader societal impact. Or, you may describe the motivation from a personal learning perspective.**
  - This project is motivated by both its real-world environmental relevance and its value as a machine learning learning experience. Monitoring biodiversity is important for understanding ecosystem health and evaluating whether ecological restoration efforts are working. However, traditional field surveys that rely on human observers are expensive, time-consuming, and difficult to scale across large habitats. By using passive acoustic monitoring data and machine learning, we can build systems that help identify species from their sounds more efficiently and across broader geographic areas. From a technical perspective, this project is also a strong opportunity to apply image classification methods in a meaningful way by converting audio recordings into spectrogram images and training a CNN to recognize species-specific acoustic patterns. It allows us to work with real-world challenges such as noisy data, limited labeled examples, and class imbalance, which are common in practical ML applications. Overall, this project combines environmental impact with hands-on experience in deep learning, signal processing, and model evaluation.

Evaluation:

- **What would the successful outcome of your project look like? In other words, under which circumstances would you consider your project to be "successful?"**
  - We will consider our project to be successful if we are able to successfully train reliable classifiers with limited labeled data. This means the model shows reliable detection for at least a subset of species.Our goal is to achieve at least a 70%-80% accuracy in classifying the species.
  - In addition, we would like to develop a preprocessing technique that involves noise reduction prior to feature extraction.
- **How do you measure success, specific to this project, from a technical standpoint?**
  - Technically, success will be measured using: Macro-F1 score, Precision and Recall, Confusion matrix evaluation.
  - The evaluation of noise reduction will be assessed by the impact on the classification performance and the clarity achieved in the spectrogram images. A successful outcome would mean an improvement in signal quality and an enhanced model accuracy for clips with noisy backgrounds.

Resources:

- **What resources are you going to use (datasets, computer hardware, computational tools,)?**
  - We are going to employ a dataset of bird sounds for animal classification off Kaggle. We plan to use a program called Audioalter or the Python library Librosa to convert all the sounds into visual representation through a spectrogram. We can then classify the images this way.
  - As far as computational tools, we will be using google colab to leverage the use of GPU's. This will allow us to train efficient deep learning models such as CNN on the spectrogram images. Additional tools we may use include Python libraries like Tensorflow for developing the model and for training.

Contributions:

Clearly indicate what computational and writing task each member of your group will be participating:

All three members will collaborate on the model evaluation, error analysis, and interpretation of results. The preprocessing and model training tasks will be divided among the members. The report will be written by all members and we will evenly distribute the work.

Works Cited

Hawkins, Joseph E. "Human Ear - Bone Conduction, Hearing, Vibration | Britannica." *Britannica*, www.britannica.com/science/ear/Transmission-of-sound-by-bone-conduction. Accessed 27 Feb. 2026.

Youssef K, Barakat JMH, El Mir G, Said S, Kork SA, Eleyan A. Animal Species Classification from Vocalizations Using Cochlear-Inspired Audio Features and Machine Learning. *Biomimetics*. 2025; 10(12):830. https://doi.org/10.3390/biomimetics10120830